

大环内酯类抗菌药物介导的药物相互作用临床试验文献评估

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摘要 目的:评估大环内酯类抗菌药物红霉素、克拉霉素与阿奇霉素介导的药物相互作用,为临床合理用药提供参考。方法:检索PubMed、中国知网(CNKI)、万方等数据库中关于红霉素、克拉霉素与阿奇霉素介导的药物相互作用的临床试验文献,并搜集其药品说明书,采用美国食品与药物管理局(FDA)推荐的药物相互作用标准进行统计分析。结果:临床试验发现有20种药物与红霉素存在药物相互作用可能性,有强、中度和弱相互作用的分别为3、6、11种;22种药物与克拉霉素存在药物相互作用可能性,有强、中度或弱相互作用的分别为2、11、9种;5种药物与阿奇霉素存在药物相互作用可能性,且全部为弱相互作用。以咪达唑仑为CYP3A4底物评估三者介导的药物相互作用强度,发现克拉霉素、红霉素和阿奇霉素分别引起强、中度和弱相互作用。结论:红霉素和克拉霉素发生药物相互作用的可能性较阿奇霉素大,可能原因是红霉素和克拉霉素对CYP3A4抑制强度远大于阿奇霉素。
关键词 药物相互作用;CYP3A4;红霉素;克拉霉素;阿奇霉素

大环内酯类(Macrolides)是一类具有14、15、16元大环内酯环基本结构与相似抗菌谱的抗菌药物。大环内酯类抗菌药物常用于治疗大多数革兰氏阳性菌、部分革兰氏阴性菌(如奈瑟菌、军团菌、嗜血杆菌等)及非典型病原体感染。临床上大多数药物相互作用是由药物代谢酶或药物转运体介导^[1-4]。大环内酯类抗菌药物为主要的药物代谢酶CYP3A4与药物转运体P糖蛋白(P-glycoprotein, P-GP)抑制剂,且不同大环内酯类药物对上述药物代谢酶和转运体的抑制强度有所差别,故不同大环内酯类药物介导的药物相互作用是否亦有差别尚不清楚。因此,本研究采用检索临床文献方法来评估常用大环内酯类抗菌药物红霉素(Erythromycin)、克拉霉素(Clarithromycin)与阿奇霉素(Azithromycin)介导的药物相互作用,为临床合理用药提供参考。

1 资料与方法

1.1 资料来源

考察红霉素、克拉霉素与阿奇霉素所介导的药物相互作用的曲线下面积比值(AUCR=AUC_{有抑制剂}/AUC_{无抑制剂})。纳入标准:红霉素、克拉霉素与阿奇霉素介导的药物相互作用的临床研究,且有AUCR报告。排除标准:红霉素、克拉霉素与阿奇霉素介导的药物相互作用的非临床研究,或者无AUCR报告的临床研究。所有数据来源于PubMed、中国知网(CNKI)、万方等数据库及美国食品与药物管理局(FDA)网站。

1.2 方法

在CNKI、万方等国内数据库采用“红霉素”“克拉霉素”“阿奇霉素”与“曲线下面积”“AUC”等检索词检索药物相互作用有关文献。在PubMed数据库则采用“Erythromycin”“Clarithromycin”“Azithromycin”与“AUC”“area under curve”“Area under curves”“Area under the concentration-time curve”“Area under the plasma concentration-time curve”“Area under the serum concentration time curves”等检索词检索文献。在FDA网站(www.FDA.org)检索红霉素、克拉霉素与阿奇霉素的药品说明书。检索时间限定为数据库建库到2016年6月10日。首先排除非临床试验与无法提供AUC的文献,保留临床试验文献。然后逐一阅读文献与药品说明书,提取剂量、底物、AUCR等数据。如有相同底物被同一抑制剂的药物相互作用被多篇文献报道,则估算AUCR平均值纳入最终分析。药物相互作用风险程度评估采用美国FDA推荐标准——弱相互作用:1.25≤AUCR<2.0;中度相互作用:2.0≤AUCR<5.0;强相互作用:AUCR≥5.0。

2 结果

2.1 红霉素介导的药物相互作用

红霉素介导的药物相互作用见表1。如表1所示,临床试验共测试红霉素与35种药物的相互作用,其中利多卡因又分口服和静脉两种给药途径,共计有36个药物相互作用结果。研究发现与红霉素可能有强相互作用的3种药物为双氢麦角隐亭、辛伐他汀和丁螺环酮;有中度相互作用的6种药物为依维莫司、咪达唑仑、乌利司

803-806.
[42] 黎颖然,卢荣枝.利奈唑胺相关新生儿血小板减少4例
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他、西地那非、利多卡因(口服)和非洛地平;有弱相互作用的11种药物为沙奎那韦、佐匹克隆、希美加群、罗氟司特、非索非那定、伏立康唑、氯雷他定、阿托伐他汀、雌二醇、洛沙坦和喹硫平。从药物类别来看,红霉素与羟甲基戊二酰辅酶A(HMG-CoA)还原酶抑制剂或H₁受体拮抗药的相互作用研究较多,分别有4种药物。

表1 红霉素介导的药物相互作用

红霉素剂量, g/d	底物	类别	AUCR	文献第一作者(发表年份)
1.5	双氢麦角隐亭	多巴胺受体激动药	16.50	de Mey C(2001) ^[5]
1.5	辛伐他汀	HMG-CoA还原酶抑制剂	6.20	Kantola T(1998) ^[6]
1.5	丁螺环酮	镇静催眠药	6.00	Kivisto KT(1997) ^[7]
1.5	依维莫司	抗肿瘤药	4.40	Kovarik JM(2005) ^[8]
1.5	咪达唑仑	镇静催眠药	3.81	Zimmermann T(1996) ^[9]
1	乌利司他	避孕药	3.27	Pohl O(2013) ^[10]
1	西地那非	性功能改善药	2.80	Muirhead GJ(2002) ^[11]
1.5	利多卡因(口服)	局部麻醉药	2.57	Isohanni MH(2006,1999) ^[12-13]
1	非洛地平	降压药	2.49	Bailey DG(1996) ^[14]
1	沙奎那韦	抗人类免疫缺陷病毒(HIV)	1.99	Grub S(2001) ^[15]
1.5	佐匹克隆	镇静催眠药	1.80	Aranko K(1994) ^[16]
0.5	希美加群	抗凝药	1.78	Matsson EM(2011) ^[17]
1.5	罗氟司特	磷酸二酯酶(PDE)抑制剂	1.70	Lahu G(2009) ^[18]
0.25	非索非那定	H ₁ 受体拮抗药	1.61	Petri N(2006) ^[19]
1.5	伏立康唑	抗真菌药	1.51	Shi HY(2010) ^[20]
1.5	氯雷他定	H ₁ 受体拮抗药	1.40	Brannan MD(1995) ^[21]
2	阿托伐他汀	HMG-CoA还原酶抑制剂	1.33	Siedlik PH(1999) ^[22]
1.5	雌二醇	避孕药	1.33	Blode H(2012) ^[23]
2	氯沙坦	降压药	1.32	Williamson KM(1998) ^[24]
1.5	喹硫平	抗精神病药	1.29	Li KY(2005) ^[25]
1.5	西立伐他汀	HMG-CoA还原酶抑制剂	1.21	Mück W(1998) ^[26]
1.5	罗哌卡因	局部麻醉药	1.21	Jokinen MJ(2000) ^[27]
2	瑞舒伐他汀	HMG-CoA还原酶抑制剂	1.20	Cooper KJ(2003) ^[28]
0.5	环索奈德	糖皮质激素	1.16	Nave R(2005) ^[29]
1.5	地西洋	镇静催眠药	1.15	Luurila H(1996) ^[30]
1	舍吲哚	抗精神病药	1.13	Wong SL(1997) ^[31]
1.5	利多卡因(静脉)	局部麻醉药	1.11	Isohanni MH(1998) ^[32]
1.5	地氯雷他定	H ₁ 受体拮抗药	1.10	Banfield C(2002) ^[33]
0.5	茶碱	支气管舒张剂	1.06	Paulsen O(1987) ^[34]
0.333	左卡巴斯汀	H ₁ 受体拮抗药	1.03	Pesco-Koplowitz L(1999) ^[35]
0.999	非尔氨酯	抗癫痫药	1.03	Sachdeo RC(1998) ^[36]
1	奥卡西平	抗癫痫药	1.02	Keränen T(1992) ^[37]
2	伏立康唑	抗真菌药	1.01	Purkins L(2003) ^[38]
2	阿加曲班	抗凝药	1.00	Tran JQ(1999) ^[39]
1	噻加宾	抗癫痫药	0.95	Thomsen MS(1998) ^[40]
1.5	氯氮平	抗精神病药	0.88	Hagg S(1999) ^[41]

2.2 克拉霉素介导的药物相互作用

克拉霉素介导的药物相互作用见表2。如表2所示,临床试验共测试克拉霉素与36种药物的相互作用。研究发现与克拉霉素可能有强相互作用的2种药物为辛伐他汀和咪达唑仑;有中度相互作用的11种药物为秋水仙碱、他克莫司、西布曲明、卡麦角林、氯胺酮、阿托伐他汀、孟鲁司特、西地那非、奥美拉唑、普伐他汀和羟考酮;有弱相互作用的9种药物:曲唑酮、利福布丁、利奈唑胺、埃索美拉唑、兰索拉唑、地高辛、达比加群、茚地那韦和安贝生坦。从药物类别来看,克拉霉素与质子泵抑制剂、抗HIV药或HMG-CoA还原酶抑制剂的相互作用研究较多,分别有6、4、3种药物。

表2 克拉霉素介导的药物相互作用

克拉霉素 剂量,g/d	底物	类别	AUCR	文献第一作者(发表年份)
1	辛伐他汀	HMG-CoA还原酶抑制剂	9.85	Jacobson TA(2004) ^[42]
1	咪达唑仑	镇静催眠药	6.50	Greenblatt DJ(2015) ^[43]
0.5	秋水仙碱	抗痛风药	3.82	Davis MW(2013) ^[44]
1	他克莫司	免疫抑制药	3.69	Kunicki PK(2005) ^[45]
0.5	西布曲明	减肥药	3.35	Pan W(2013) ^[46]
0.4	卡麦角林	抗帕金森药	2.62	Nakatsuka A(2006) ^[47]
1	氯胺酮	全身麻醉药	2.60	Hagelberg NM(2010) ^[48]
1	阿托伐他汀	HMG-CoA还原酶抑制剂	2.54	Jacobson TA(2004) ^[42] Amsden GW(2002) ^[49] Shin J(2011) ^[50]
1	孟鲁司特	支气管舒张药	2.44	Hegazy SK(2012) ^[51]
0.5	西地那非	性功能改善药	2.28	Hedaya MA(2006) ^[52]
0.8	奥美拉唑	质子泵抑制剂	2.11	Furuta T(1999) ^[53] Calabresi L(2004) ^[54]
1	普伐他汀	HMG-CoA还原酶抑制剂	2.10	Jacobson TA(2004) ^[42]
1	羟考酮	镇痛药	2.00	Liukas A(2011) ^[55]
1	曲唑酮	抗精神病药	1.99	Farkas D(2009) ^[56]
1	利福布丁	抗结核药	1.99	Hafner R(1998) ^[57]
0.5	利奈唑胺	抗菌药	1.85	Bolhuis MS(2013) ^[58]
1	埃索美拉唑	质子泵抑制剂	1.70	Hassan-Alin M(2006) ^[59]
1	兰索拉唑	质子泵抑制剂	1.70	Saito M(2005) ^[60]
0.5	地高辛	抗慢性心力衰竭药	1.70	Rengelshausen J(2003) ^[61]
1	达比加群	抗凝药	1.49	Delavenne X(2013) ^[62]
1	茚地那韦	抗HIV药	1.47	Boruchoff SE(2000) ^[63]
1	安贝生坦	降肺动脉高压药	1.41	Markert C(2013) ^[64]
0.5	罗哌卡因	局部麻醉药	1.20	Jokinen MJ(2001) ^[65]
1	安普那韦	抗HIV药	1.18	Brophy DF(2000) ^[66]
1	咖啡因	中枢兴奋药	1.14	Bruce MA(2001) ^[67]
1	利托那韦	抗HIV药	1.13	Ouellet D(1998) ^[68]
1	泮托拉唑	质子泵抑制剂	1.12	Calabresi L(2004) ^[54]
0.4	舒马曲坦	抗偏头痛药	1.07	Moore KH(2002) ^[69]
1	磺胺甲噁唑	抗菌药	1.07	Winter HR(2004) ^[70]
0.4	普仑司特	支气管舒张药	1.06	Nakade S(2008) ^[71]
1	甲苯磺丁脲	降糖药	1.06	Bruce MA(2001) ^[67]
1	唑吡坦	镇静催眠药	1.04	Farkas D(2009) ^[56]
0.8	雷贝拉唑	质子泵抑制剂	1.00	Shimizu M(2006) ^[72]
1	氨苯砜	抗麻风药	1.00	Winter HR(2004) ^[73]
1	达芦那韦	抗HIV药	0.89	Sekar VJ(2008) ^[74]
0.5	艾普拉唑	质子泵抑制剂	0.86	Cao S(2012) ^[75]

2.3 阿奇霉素介导的药物相互作用

阿奇霉素介导的药物相互作用见表3。如表3所示,临床试验共测试阿奇霉素与22种药物的相互作用。研究发现,与阿奇霉素可能有弱相互作用的5种药物为非索非那定、希美加群、秋水仙碱、伊维菌素和咪达唑仑;无强或中度相互作用的药物。从药物类别来看,阿奇霉素与H₁受体阻滞药或抗HIV药的相互作用研究较多,分别有4、3种药物。

2.4 3种药物介导的药物相互作用类别比较

3种药物介导的药物相互作用类别比较见表4。如表4所示,红霉素与克拉霉素引起的药物相互作用远多于阿奇霉素,分别为20、22、5个。药物相互作用中,阿奇霉素仅有弱相互作用的临床报道,而红霉素和克拉霉素3种强度的相互作用均有报道。选用咪达唑仑(Midazolam)为CYP3A4底物来分析三者引起的药物相互作用强度,克拉霉素、红霉素与阿奇霉素分别引起强、中度、弱

表3 阿奇霉素介导的药物相互作用

阿奇霉素 剂量,g/d	底物	类别	AUCR	文献第一作者(发表年份)
0.5	非索非那定	H ₂ 受体阻滞药	1.67	Gupta S(2001) ^[76]
0.5	希美加群	抗凝药	1.60	Dorani H(2007) ^[77]
0.5	秋水仙碱	抗痛风药	1.57	Davis MW(2013) ^[44]
0.5	伊维菌素	抗寄生虫药	1.31	Amsden GW(2007) ^[78]
0.5	咪达唑仑	镇静催眠药	1.26	Zimmermann T(1996) ^[79]
1.2	去羟肌苷	抗HIV药	1.14	drug label information
1.2	齐多夫定	抗HIV药	1.12	drug label information
0.5	伏立康唑	抗真菌药	1.08	Purkins L(2003) ^[80]
0.5	环孢素A	免疫抑制剂	1.07	Bachmann K(2003) ^[79]
0.5	西地那非	性功能改善药	1.05	Muirhead GJ(2002) ^[81]
0.5	地氯雷他定	H ₂ 受体阻滞药	1.05	Gupta S(2001) ^[76]
0.5	茶碱	支气管舒张剂	1.05	drug label information
0.5	卢帕他定	H ₂ 受体阻滞药	1.03	Solans A(2008) ^[80]
0.5	阿苯达唑	抗寄生虫药	1.03	Amsden GW(2007) ^[78]
0.5	西替利嗪	H ₂ 受体阻滞药	1.02	drug label information
0.5	三唑仑	镇静催眠药	1.02	drug label information
1.2	氟康唑	抗真菌药	1.01	drug label information
0.5	阿托伐他汀	HMG-CoA还原酶抑制剂	0.98	Amsden GW(2002) ^[80]
1.2	磺胺甲噁唑	抗菌药物	0.96	drug label information
0.5	卡马西平	抗癫痫药	0.96	drug label information
1	氯喹	抗寄生虫药	0.95	Cook JA(2006) ^[81]
0.6	依法韦仑	抗HIV药	0.95	drug label information

相互作用(AUCR分别为6.50、3.81、1.26),见图1A。由此可见,克拉霉素为CYP3A4的强抑制剂,红霉素为CYP3A4的中度抑制剂,而阿奇霉素为CYP3A4的弱抑制剂。选用地高辛或非索非那定为P-GP底物来分析三者引起的药物相互作用强度,克拉霉素、红霉素与阿奇霉素都导致弱相互作用(AUCR分别为1.70、1.61、1.67),表明三者对P-GP的抑制强度可能类似,见图1B。

表4 3种药物介导的药物相互作用类别比较

药物相互作用类别	红霉素	克拉霉素	阿奇霉素
强(AUCR≥5)	3	2	0
中度(2.0≤AUCR<5.0)	6	11	0
弱(1.25≤AUCR<2.0)	11	9	5
合计	20	22	5

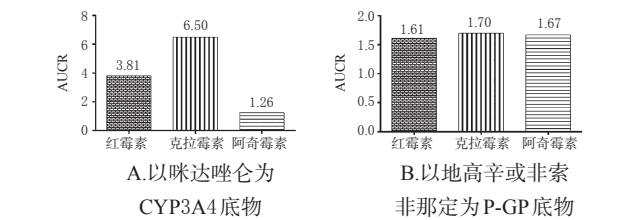


图1 红霉素、克拉霉素和阿奇霉素介导的药物相互作用强度比较

3 讨论

本研究结果表明,红霉素和克拉霉素发生药物相互作用的可能性与强度远大于阿奇霉素。可能原因是三者对药物主要代谢酶CYP3A4活性而非药物转运体P-GP活性的抑制强度不同造成的。以咪达唑仑为CYP3A4底物评估三者介导的药物相互作用强度,发现克拉霉素、红霉素和阿奇霉素分别引起强、中度和弱相互作用,因此克拉霉素、红霉素和阿奇霉素分别是

CYP3A4的强抑制剂、中度抑制剂和弱抑制剂^[9, 43-44]。而以地高辛或非索非那定为P-GP底物评估三者介导的药物相互作用强度,发现克拉霉素、红霉素和阿奇霉素都引起弱相互作用,因此3种药物对P-GP的转运活性抑制程度可能相当^[19, 61, 76]。

从药物类别来看,研究者主要关注3种大环内酯类抗菌药物与HMG-CoA还原酶抑制剂、质子泵抑制剂、H₂受体拮抗药和抗HIV药的药物相互作用。以HMG-CoA还原酶抑制剂为例,红霉素、克拉霉素和阿奇霉素介导的药物相互作用强度还是有所区别。以CYP3A4底物阿托伐他汀为例,克拉霉素、红霉素和阿奇霉素分别介导的阿托伐他汀的AUCR为2.54、1.33和0.98,表明三者分别引起中度、弱和无相互作用^[22, 42, 49-50]。上述差异可能是由于克拉霉素、红霉素和阿奇霉素对CYP3A4活性抑制程度不同造成的。此外,同一大环内酯类抗菌药物对不同HMG-CoA还原酶抑制剂的药物相互作用可能性亦有区别。红霉素介导的辛伐他汀、阿托伐他汀、西立伐他汀(已退市)和瑞舒伐他汀的AUCR分别为6.20、1.33、1.21、1.20,因此红霉素仅与辛伐他汀和阿托伐他汀存在强与弱药物相互作用可能性^[6, 22, 26, 28, 42, 50]。同样,克拉霉素介导的辛伐他汀、阿托伐他汀和普伐他汀的AUCR分别为9.85、2.54、2.10,故克拉霉素与辛伐他汀存在强相互作用可能性,而与阿托伐他汀和普伐他汀存在中度相互作用可能性^[42, 82]。造成上述药物相互作用差异的主要原因是不同HMG-CoA还原酶抑制剂经CYP3A4的代谢程度有差异。辛伐他汀和阿托伐他汀主要经CYP3A4代谢,而普伐他汀和瑞舒伐他汀则少量经CYP3A4代谢。故红霉素和克拉霉素与辛伐他汀和阿托伐他汀有明显的药物相互作用可能性。因此,对于伴有胃十二指肠溃疡的高胆固醇血症患者,如选用含克拉霉素的幽门螺杆菌根治方案,则降胆固醇药应尽量避免使用辛伐他汀和阿托伐他汀。

综上所述,红霉素和克拉霉素与其他药物发生相互作用的可能性与强度远大于阿奇霉素,可能原因是红霉素和克拉霉素对CYP3A4抑制强度远大于阿奇霉素。其次,阿奇霉素是大环内酯类抗菌药物中发生药物相互作用较少的品种。

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